

VEGA

Runbook Following Agent

Verify · Execute · Guide · Audit

Team: Hack Infinite

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Problem Statement

Operational runbooks are still executed manually.

Manual Execution

DevOps teams run multi-step runbooks by hand — slow, error-prone and inconsistent across operators.

No Risk Awareness

There is no automatic check for whether a command is safe, medium or high risk before it executes.

Missing Audit Trail

Outcomes, approvals and errors are rarely captured in one place, making incident review difficult.

No Auto-Recovery

When a step fails, recovery depends entirely on human intuition; there is no AI-assisted remediation loop.

Solution — VEGA

An AI-powered runbook execution engine with risk classification, human-in-the-loop approval, and automatic error remediation.



Verify

Parses runbooks and classifies every command as SAFE, MEDIUM, or HIGH using a local Ollama LLM.



Execute

Runs SHELL, REST_API, DB_QUERY, and CLOUD_CLI steps through a unified, allowlist-enforced executor.



Guide

Pauses on risky steps for operator approval and sends real-time Discord notifications.



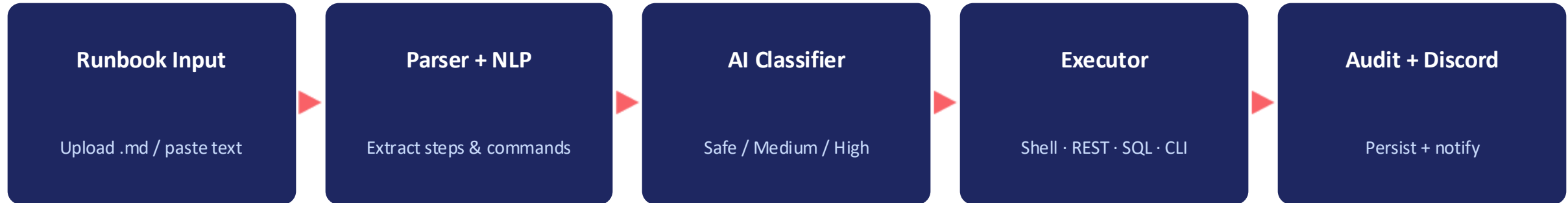
Audit

Persists every run and step in SQLite/MySQL with a structured audit report per execution.

Tech Stack

Frontend	React 19 • Vite 8 • Plain CSS
Backend	Python 3 • Flask 3 • Flask-CORS
Database	SQLAlchemy 2 • SQLite (default) / MySQL
AI Runtime	Ollama (local LLM, llama3 model)
Auto-Remediation	MCP SDK (Model Context Protocol)
Notifications	Discord Webhooks
File Parsing	pypdf • python-docx • openpyxl • markdown

Architecture & Workflow



Human-in-the-Loop & Auto-Remediation

- Medium / High risk steps pause and wait for explicit operator approval.
- A configurable allowlist provides a hard security boundary independent of AI.
- Failed steps trigger the MCP client to fetch a corrected command and auto-retry.
- Every run and step is persisted with status, duration, and operator decisions.

Key Features

1

Runbook Parsing

Markdown upload or paste; NLP infers commands from natural language.

2

Multi-Type Execution

SHELL, REST_API, DB_QUERY (SQL), and CLOUD_CLI (aws, gcloud, kubectl, terraform).

3

AI Risk Classification

Ollama LLM with deterministic rule-based fallback when offline.

4

Approval Workflow

Risky steps pause for explicit approval before continuing.

5

Allowlist Enforcement

Configurable safe-command allowlist independent of AI output.

6

MCP Auto-Remediation

On failure, retrieves a corrected command and retries automatically.

Thank You

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